Probability Theory

Lecture Notes

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Probability Theory

Lecture Notes

Fall 2026

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Lecture 01

Introduction to Probability

Date: Wednesday, 05 August 2026
Instructor: Dr. Nishant Chandgotia

1.1 Motivation and Some Problems

Question 1. Sleeping Beauty Problem

A princess is put into a deep sleep by a powerful drug administered by a villain. The drug ensures that she will remain asleep for two years, except for possible brief awakenings determined by the outcome of a fair coin toss.
The villain tosses a fair coin.

- If the coin lands **Heads**, the princess is awakened *once*, briefly, at the end of the **first year**. After this awakening, her memory of having awakened is completely erased, and she is put back to sleep until the end of the second year.
- If the coin lands **Tails**, the princess is not awakened after the first year. Instead, she is awakened *once*, briefly, at the end of the **second year**.

Whenever the princess awakens, she has no information about the current date or the passage of time. Furthermore, if she had awakened previously, all memory of that awakening has been erased. Consequently, upon waking, she cannot distinguish between the two possible awakening scenarios.
Suppose that the princess wakes up and is immediately asked the following question:

"What probability should you assign to the event that the coin landed Heads?"

What should the princess answer?

Solution:

1. $\frac{1}{2}$ No information exchanged so probability should be $\frac{1}{2}$, If repeated indipendently a milliion times, half chances each.
2. $\frac{2}{3}$ There are possibilities of this. Proportion of waking during Heads is $\frac{2}{3}$,and $\ldots\frac{1}{3}$
3. $\frac{3}{4}$

Villain tosses once, H $\rightarrow$Villain tosses again, Two Possibilities H and T, so,

- HH $\rightarrow \frac{1}{4}$
- HT $\rightarrow \frac{1}{4}$

- T $\rightarrow \frac{1}{2}$

Important

All the solutions above are wrong. The Above problem has no solution, Because it is ill posed, and this ill posedness is with respect to current Framework.

Question 2. Monty Hall Problem

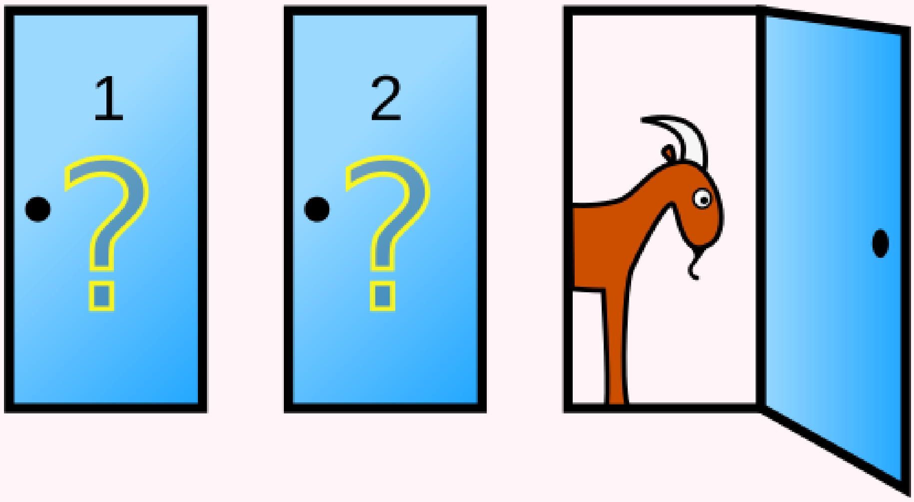

Figure 1: Monty Hall Problem

There are three doors: one hides a car and the other two hide goats. After you choose one door say A, the host opens another door revealing a goat.

1. Question: Should you switch?
2. Probability car behind C > Probability car behind A.

Solution: Yes, we should change our decision,

Initially, the probability that your chosen door contains the car is $P(\text{chosen door}) = \frac{1}{3}$, while the probability that the car is behind one of the other two doors is $P(\text{other two doors}) = \frac{2}{3}$.

Since the host always reveals a goat, this entire probability concentrates on the remaining unopened door. Therefore, $P(\text{Win by staying}) = \frac{1}{3}, \qquad P(\text{Win by switching}) = \frac{2}{3}$. Hence, switching gives twice the probability of winning compared to staying.

Remark

The Above question (Monty Hall Problem) is not ill posed.

Summary

- Sleeping Beauty Problem
- Monty Hall Problem

End of Lecture

Probability Theory

Lecture 02

Introduction to Probability

Date: Friday, 07 August 2026
Instructor: Dr. Nishant Chandgotia

2.1 Some Introductory Idea

Suppose we toss a fair coin n-times. $HTH\ HTTH\ \cdots\ T$

What is the Probability of the above sequence ? Its- $\sim \frac{1}{2^n}$

If we have all sequences with k-heads then $P(k\text{-heads}) = \frac{\binom{n}{k}}{2^n}$, and it will be maximum when $k = \frac{n}{2}$.

If $x_1, x_2, \ldots, x_k$ are randomly independently chosen, then $\sum_{i=1}^{k} x_i \sim \frac{r+1}{2} \cdot k$, if they are chosen from the uniform distribution on $\{1, 2, \ldots, r\}$.

Now Stirling's Formula, $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$, for very large n.

Now,

$$\frac{P(\text{number of heads } = k)}{P(\text{number of heads } = k+1)} = \frac{k+1}{n-k}.$$

So, $\frac{k+1}{n-k} > 1 \iff k > \frac{n-1}{2}$, and $\frac{k+1}{n-k} < 1 \iff k < \frac{n-1}{2}$.

As $\binom{n}{k}$ is maximum when $k = \frac{n}{2}$, therefore, $\frac{\binom{n}{n/2}}{2^n} \sim \frac{2}{\sqrt{2\pi n}}$ (By using Stirling's Formula). It would be unjustified to say #Heads $= \frac{n}{2}$ with high probability.

Suppose $k = \alpha n$, $\alpha \neq \frac{1}{2}$.

Then $\frac{\binom{n}{k}}{2^n} \sim \frac{1}{\sqrt{2\pi n}} e^{n(-\alpha \log \alpha - (1-\alpha)\log(1-\alpha) - \log 2)}$.

Let, $H(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha)$, be Boltzman Shanon Entropy (Mathematical theory of Communications.

Hence,

$$\frac{\binom{n}{k}}{2^n} \sim \frac{1}{\sqrt{2\pi n}} e^{n\left(H(\alpha) - H\left(\frac{1}{2}\right)\right)}.$$

Now, $H(\alpha) \leq H\left(\frac{1}{2}\right)$, by using Jensen's inequality as $\log$ is a concave function.

Equivalently, $-\alpha \log \alpha - (1-\alpha)\log(1-\alpha) < \ln 2$.

Therefore, $\frac{\binom{n}{k}}{2^n} \longrightarrow 0$ exponentially.

Now,

$$P(\#\text{Heads} \in ((\alpha-\varepsilon)n, (\alpha+\varepsilon)n)) = \sum_{k=(\alpha-\varepsilon)n}^{(\alpha+\varepsilon)n} \frac{\binom{n}{k}}{2^n}. \qquad \text{, where} \varepsilon > 0, \qquad \frac{1}{2} \notin (\alpha-\varepsilon, \alpha+\varepsilon).$$

Also,

$$\frac{2n\varepsilon}{2^n}\binom{n}{(\alpha-\varepsilon)n} \leq \cdots \leq \frac{2n\varepsilon}{2^n}\binom{n}{(\alpha+\varepsilon)n}.$$

Therefore, the asymptotic bounds are $\frac{2n\varepsilon}{\sqrt{2\pi n}} e^{n\left(H(\alpha-\varepsilon)-H\left(\frac{1}{2}\right)\right)}$ and $\frac{2n\varepsilon}{\sqrt{2\pi n}} e^{n\left(H(\alpha+\varepsilon)-H\left(\frac{1}{2}\right)\right)}$.

Therefore, $P(\#\text{Heads} \in ((\alpha-\varepsilon)n, (\alpha+\varepsilon)n)) \longrightarrow 0$ exponentially as $n \to \infty$.

Hence,

$$P\left(\#\text{Heads} \in n\left(\frac{1}{2}-\varepsilon, \frac{1}{2}+\varepsilon\right)\right) \longrightarrow 1.$$

This is called the **Large Deviation Phenomenon**.

$P(\text{staying close to mean}) \longrightarrow 1$,

whereas $P(\text{deviating from the mean significantly}) \longrightarrow 0$ exponentially.

In our case, the rate function is $H(\alpha) - H\left(\frac{1}{2}\right)$.

Let $f_n = \frac{\#\text{Heads}}{n}$. Then $f_n \longrightarrow f \qquad$ in measure.

Therefore,

$$P\left(\left|\frac{\#\text{Heads}}{n} - \frac{1}{2}\right| > \varepsilon\right) \longrightarrow 0 \qquad \text{as } n \to \infty. \tag{1}$$

We call this **convergence in probability**.

Later we shall use the **Borel–Cantelli Lemma** to prove from (1) that $\frac{\#\text{Heads}}{n} \longrightarrow \frac{1}{2}$ almost surely.

The sample space is $\Omega = \{H, T\}^{\mathbb{N}}$. We equip Ω with the product topology, where each copy of

$\{H, T\}$ carries the discrete topology. The corresponding Borel σ-algebra, $\mathcal{F} = \mathcal{B}(\Omega)$, is generated by the cylinder sets. The basis of the product topology consists of the cylinder sets, namely sets of the form $\{\omega \in \Omega : \omega_{i_1} = a_1, \ldots, \omega_{i_k} = a_k\}$, where $i_1, \ldots, i_k$ are finitely many coordinates and each $a_j \in \{H, T\}$.

If we only specify the first n coordinates, they look like $\{\omega : \omega_1 = a_1, \ldots, \omega_n = a_n\}$, which are the basic cylinders.

The complete probability space is $(\Omega, \mathcal{F}, \mu) = \left(\{H, T\}^{\mathbb{N}}, \mathcal{B}\left(\{H, T\}^{\mathbb{N}}\right), \mu\right)$,

Recall,

$$P\left(\#\text{Heads} = \frac{n}{2}\right) \sim \frac{2}{\sqrt{2\pi n}}.$$

Suppose

$$c > 0,$$

and let $\left|k - \frac{n}{2}\right| < c\sqrt{n}$.

Then

$$P(\#\text{Heads} = k) = \frac{\frac{n!}{(n-k)!k!}}{2^n}.$$

Using Stirling's formula,

$$P(\#\text{Heads} = k) \sim \frac{1}{\sqrt{\pi n/2}} \cdot \frac{1}{\left(2 - \frac{2k}{n}\right)^{n-k} \left(\frac{2k}{n}\right)^k}.$$

$$= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \cdot \frac{1}{\left(1 + \left(1 - \frac{2k}{n}\right)\right)^{n-k} \left(1 + \left(\frac{2k}{n} - 1\right)\right)^k}.$$

Hence,

$$= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \exp\left(-\frac{(n-2k)^2}{n}\right).$$

Equivalently, $= \frac{1}{\sqrt{2\pi \cdot \frac{n}{4}}} \exp\left(-\frac{\left(k - \frac{n}{2}\right)^2}{2 \cdot \frac{n}{4}}\right)$, which resembles the normal distribution.

$$\mu = \frac{1}{2}, \qquad \sigma = \frac{1}{2}.$$

Hence,

$$= \frac{1}{\sqrt{2\pi\sigma^2 n}} \exp\left(-\frac{(k - \mu n)^2}{2\sigma^2 n}\right).$$

We can prove, $P\left(a < \frac{\#\text{Heads} - \mu n}{\sigma\sqrt{n}} < b\right) \longrightarrow \int_a^b \varphi(x)\, dx, \qquad \text{as } n \to \infty$. (**Central Limit Theorem / De Moivre–Laplace**)

End of Lecture

2.2 Quiz, 10 August 2026

Quiz 1

1. Let $p < \frac{1}{2}$. Suppose we toss n coins independently such that the probability of heads is p and tails is $q = 1 - p$.

   For what value of k is the probability of getting k heads maximised?

2. For probability vectors $P = (p_1, p_2, \ldots, p_n), \qquad Q = (q_1, q_2, \ldots, q_n)$, where $p_i, q_i > 0, \qquad \sum_i p_i = \sum_i q_i = 1$, define the Kullback–Leibler divergence by

   $$D_{KL}(P \| Q) = \sum_i p_i \log\left(\frac{p_i}{q_i}\right).$$

   Use Jensen's inequality to show that $D_{KL}(P\|Q) \geq 0$ with equality if and only if $P = Q$.

3. Let $\alpha \in (0, 1)$. Prove that the number of heads equalling $n\alpha$ is asymptotically

   $$\frac{1}{\sqrt{2\pi n\alpha(1-\alpha)}} \exp\left(-nD_{KL}(P\|Q)\right),$$

   where $P = (\alpha, 1-\alpha), \qquad Q = (p, q)$.

Probability Theory

Lecture 03

Introduction to Probability

Date:	Wednesday, 12 August 2026
Instructor:	Dr. Nishant Chandgotia

3.1 Central Limit Theorem and Convergence

Suppose we toss a fair coin independently n times. The Central Limit Theorem suggests that

$$P\left(a < \frac{\#\text{Heads} - \mu n}{\sigma\sqrt{n}} < b\right) \longrightarrow \int_a^b \varphi(x)\,dx, \qquad n \to \infty,$$

where, for a fair coin, and
$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$.

3.1.1 Galton Board

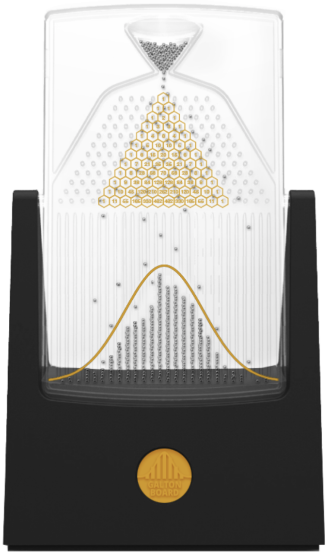

Figure 2: Galton board. Source: Wikimedia Commons.

The Galton board gives a visual representation of a process whose distribution approaches a normal distribution.

3.1.2 Random Walk

A *random walk* is a stochastic process in which the position changes randomly at each step.

For example, consider a person starting at the origin. At every step, the person tosses a fair coin:

$$\begin{cases} H & \longrightarrow \text{move one step to the right,} \\ T & \longrightarrow \text{move one step to the left.} \end{cases}$$

This gives the classical one-dimensional symmetric random walk.

There are several natural questions:

Questions

- Will the person return to the origin?
- Will the person escape to infinity?
- How often will the person return to the origin?

3.1.3 Convergence in Distribution

We know,

$$\frac{\#\text{Heads}}{n} \longrightarrow \frac{1}{2}.$$

Now define

$$f_n = \frac{\#\text{Heads} - \mu n}{\sigma\sqrt{n}}.$$

Then the Central Limit Theorem states that

$$P(a < f_n < b) \longrightarrow \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx.$$

This leads to the notion of *weak convergence*, or *convergence in distribution*.

A natural question is whether convergence in distribution implies almost-sure convergence.

Important

Convergence in distribution does not, in general, imply almost-sure convergence.

3.1.4 An Example

Consider the following random process.

Start with 1.

Toss a fair coin independently at every step:

$$\begin{cases} H & \longrightarrow \text{multiply by } 2, \\ T & \longrightarrow \text{multiply by } -2. \end{cases}$$

After n steps, define

$$f_n = \frac{\text{product after } n \text{ steps}}{2^n}.$$

Since every multiplication contributes either 2 or -2,

$$f_n \in \{-1, 1\}.$$

Moreover,

$$P(f_n = 1) = \frac{1}{2}, \qquad P(f_n = -1) = \frac{1}{2}.$$

Thus the distribution of f_n is the same for every n.

However, f_n does not converge almost surely.

Questions

- Does f_n converge almost surely?
- Does f_n converge in L^1?
- Does f_n converge in probability?

Remark

The example illustrates that convergence in distribution is a weaker notion of convergence than almost-sure convergence.

3.2 Objects in Probability

There is a close analogy between measure theory and probability theory.

Table 1: Analogy between Measure Theory and Probability

Measure Theory	**Probability**
Measure space: $(\Omega, \mathcal{B}, \mu)$	Probability space: $(\Omega, \mathcal{B}, \mathbb{P})$
σ-algebra $\mathcal{B}$: $A \in \mathcal{B}$, where $\mu(A)$ is defined.	Event: $A \in \mathcal{B}$, for which $\mathbb{P}(A)$ is defined.
Elements: $\omega \in \Omega$	Outcomes / samples: $\omega \in \Omega$
Measurable function: $$f : (\Omega, \mathcal{B}) \longrightarrow (S, \rho)$$	Random variable: $$X : (\Omega, \mathcal{B}, \mathbb{P}) \longrightarrow (S, \rho)$$
L^1-integrability: $$\int_\Omega \lvert f \rvert \, d\mu < \infty$$	Mean / expectation: $$\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}$$
L^2-norm: $$\lVert f \rVert_{L^2}^2 = \int_\Omega \lvert f \rvert^2 \, d\mu$$	Variance: $$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$
Push-forward measure: $$f_*\mu(A) = \mu\left(f^{-1}(A)\right)$$	Distribution of X: $$\nu(A) = \mathbb{P}\left(X^{-1}(A)\right)$$
Fourier transform: $$\widehat{f}(t) = \int_{\mathbb{R}} e^{itx} f(x) \, dx$$	Characteristic function: $$\Phi_X(t) = \mathbb{E}\left[e^{itX}\right]$$
Convergence in measure	Convergence in probability
Almost everywhere (a.e.)	Almost surely (a.s.)
Convergence in L^1	Convergence in mean.

3.3 Sample Space and Sigma Algebra

The first object in probability is the *sample space*.

$$\Omega = \text{the universe of all possible outcomes.}$$

For example, for a single coin toss, $\Omega = \{H, T\}$.

For a bus arriving at a non-negative time, $\Omega = [0, \infty)$.

For infinitely many coin tosses, $\Omega = \{H, T\}^{\mathbb{N}}$.

An element of this space is an infinite sequence $\omega = (\omega_1, \omega_2, \ldots)$, $\quad \omega_i \in \{H, T\}$.

3.3.1 Sigma Algebra

Let $\mathcal{B}$ denote the σ-algebra of events.

For a single coin toss, $\mathcal{B} = \{\varnothing, \{H\}, \{T\}, \{H, T\}\}$.

For the infinite product space $\{H, T\}^{\mathbb{N}}$, the relevant σ-algebra is generated by cylinder sets.

3.3.2 Cylinder Sets

Suppose we are interested only in the first finitely many tosses.

Let $F \subset \mathbb{N}$ be a finite set and let $a : F \longrightarrow \{H, T\}$ be a fixed assignment of heads and tails.

Define $[a] = \left\{\omega \in \{H, T\}^{\mathbb{N}} : \omega_i = a(i) \text{ for every } i \in F\right\}$.

The set $[a]$ specifies the outcomes of the coordinates belonging to the finite set F, while leaving all other coordinates unrestricted.

Such sets are called *cylinder sets*.

They form a basis for the product topology on $\{H, T\}^{\mathbb{N}}$.

We want to be able to assign a probability $\mathbb{P}([a])$ to every cylinder set.

For a fair coin, if $|F| = k$, then $\mathbb{P}([a]) = \frac{1}{2^k}$.

End of Lecture

Probability Theory

Lecture 04

Introduction to Probability

Date: Wednesday, 14 August 2026
Instructor: Dr. Nishant Chandgotia

4.1 Sample Spaces, Random Variables and Their Distributions

4.1.1 Sample Space

The first object in probability is the *sample space* Ω, which is the collection of all possible outcomes.

1. **Simple coin toss** $\quad \Omega = \{H, T\}$.
2. **Arrival time of a bus** Depending on the context, we may take $\Omega = [0, 1] \quad \text{or} \quad \Omega = [0, \infty)$.
3. **Infinite sequence of coin tosses** $\quad \Omega = \{H, T\}^{\mathbb{N}}$.

4.1.2 σ-Algebra

1. A σ-algebra $\mathcal{B}$ is the collection of events whose probabilities we can measure.
   For a simple coin toss, $\quad \mathcal{B} = \{\varnothing, \{H\}, \{T\}, \{H, T\}\}$.
2. For a subset of $\mathbb{R}$, we are naturally led to the *Borel σ-algebra*.
3. For the infinite product space $\{H, T\}^{\mathbb{N}}$, consider a finite set $F \subset \mathbb{N}$ and a function $a : F \longrightarrow \{H, T\}$.
   Define the corresponding cylinder set by
   $$[a] = \left\{\omega \in \{H, T\}^{\mathbb{N}} : \omega_i = a(i) \text{ for every } i \in F\right\}.$$
   Cylinder sets form the basic building blocks for the product σ-algebra.

4.1.3 Probability Measure

A probability measure is a measure $\mathbb{P} : \mathcal{B} \longrightarrow [0, 1]$ such that $\mathbb{P}(\Omega) = 1$.

1. For a fair coin, $\quad \mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}$.

2. For the uniform probability measure on $[0,1]$,

$$\mathbb{P}([a,b]) = b-a, \qquad 0 \le a \le b \le 1.$$

3. For independent coin tosses, if $[a]$ is a cylinder set determined by $|F|$ coordinates, then

$$\mathbb{P}([a]) = 2^{-|F|}.$$

4. More generally, for a coin with $\mathbb{P}(H) = p, \qquad \mathbb{P}(T) = 1-p,$
   we have

$$\mathbb{P}([a]) = p^{\#\text{Heads}}(1-p)^{\#\text{Tails}}.$$

   Question: Why does this define a probability measure on $\{H,T\}^{\mathbb{N}}$?

4.1.4 Non-measurable Sets

Important

Not every subset of $\{H,T\}^{\mathbb{N}}$ is measurable.

The construction of non-measurable sets is related to the existence of suitable equivalence relations and translation-invariant measures.

A useful analogy is the classical construction for $\mathbb{R}$.

4.1.4.1 Proof Idea on $\mathbb{R}$

The general strategy is:

1. Find a countable subgroup of $\mathbb{R}$.
2. Find a Borel measure satisfying translation invariance:

$$\mu(A+x) = \mu(A), \qquad \forall x \in \mathbb{R}, \quad A \in \mathcal{B}.$$

4.1.4.2 Analogue for the Infinite Coin-Toss Space

Consider $\quad \left(\frac{\mathbb{Z}}{2\mathbb{Z}}\right)^{\mathbb{N}}.$

One can consider the set

$$G = \left\{ x \in \left(\frac{\mathbb{Z}}{2\mathbb{Z}}\right)^{\mathbb{N}} : x(i) = 0 \text{ for all sufficiently large } i \right\}.$$

The idea is to check that G is a countable subgroup and then investigate translation invariance of the probability measure on cylinder sets.

In particular, for a cylinder set $[a]$,

$$\mathbb{P}([a]) = 2^{-|F|}.$$

4.2 Random Variables

A random variable is a measurable function from the probability space to another measurable space.

For example, for infinitely many coin tosses, $f_n : \{H,T\}^{\mathbb{N}} \longrightarrow [0,1]$, defined by

$$f_n(\omega) = \frac{1}{n}\sum_{i=1}^{n} X_i(\omega),$$

where

$$X_i(\omega) = \begin{cases} 1, & \omega_i = H, \\ 0, & \omega_i = T. \end{cases}$$

We would like to understand the limit

$$f(\omega) = \lim_{n\to\infty} f_n(\omega).$$

4.2.1 A Random Variable Detecting the Existence of a Limit

Question 1. Random Variable for Existence of Limit

Given a probability space and a sequence of random variables (f_n), define a random variable which captures whether or not $\lim_{n\to\infty} f_n(\omega)$ exists.

For a real-valued sequence, the limit exists precisely when $\limsup_{n\to\infty} f_n(\omega) = \liminf_{n\to\infty} f_n(\omega)$ and both quantities are finite.

Define

$$\widetilde{f}(\omega) = \begin{cases} \limsup\limits_{n\to\infty} f_n(\omega) - \liminf\limits_{n\to\infty} f_n(\omega), & \text{if both are finite,} \\ +\infty, & \text{otherwise.} \end{cases}$$

Then define

$$f'(\omega) = \begin{cases} 1, & \widetilde{f}(\omega) = 0, \\ 0, & \widetilde{f}(\omega) \neq 0. \end{cases}$$

Thus

$$f'(\omega) = \mathbf{1}_{\{\lim_{n\to\infty} f_n(\omega) \text{ exists as a finite real number}\}}.$$

Remark

The precise formulation and notation may differ from the presentation in Durrett.

4.3 Distribution of a Random Variable

Let $X : (\Omega, \mathcal{B}, \mathbb{P}) \longrightarrow (S, \rho)$ be a random variable.

The distribution of X is the push-forward of $\mathbb{P}$ under X: $X_*\mathbb{P}$.

More explicitly, the distribution μ_X of X is the measure on (S, ρ) defined by

$$\mu_X(A) = \mathbb{P}\left(X^{-1}(A)\right), \qquad A \in \rho.$$

It is also common to write

$$\mathbb{P}(X \in A) \coloneqq \mathbb{P}\left(X^{-1}(A)\right).$$

Consider the random variables X_1 and X_2 arising from the independent coin-toss example above.

Then X_1 and X_2 have the same distribution: $X_1 \stackrel{d}{=} X_2$.

However, they are not the same random variable.

Indeed, $X_1(\omega) \neq X_2(\omega)$ for some ω.

Also, (X_1, X_2) is itself a random variable, taking values in the appropriate product space.

Remark

Having the same distribution does not imply that two random variables are equal.
In symbols,

$$X = Y \quad \Longrightarrow \quad X \stackrel{d}{=} Y,$$

but in general,

$$X \stackrel{d}{=} Y \quad \not\Longrightarrow \quad X = Y.$$

4.4 Types of Random Variables

Random variables can be classified according to their distributions.

1. **Discrete random variables**
2. **Continuous random variables**
   (a) Absolutely continuous random variables.
   (b) Singular random variables.

- These classifications are determined by the distribution of the random variable.
- A discrete random variable has an *atomic distribution.*

4.4.1 Examples of Discrete Random Variables

1. **Bernoulli random variable** Let $p \in [0,1]$. We write $X \sim \mathrm{Ber}(p)$ if
$$\mathbb{P}(X=1)=p, \qquad \mathbb{P}(X=0)=1-p.$$
This notation means that X has the Bernoulli distribution with parameter p.

2. **Binomial random variable**
Let $n \in \mathbb{N}, \qquad p \in [0,1]$. We write $X \sim \mathrm{Bin}(n,p)$
if
$$\mathbb{P}(X=r)=\binom{n}{r}p^r(1-p)^{n-r}, \qquad r=0,1,\ldots,n.$$

3. **Poisson random variable**
Let $\lambda > 0$. We write $\quad X \sim \mathrm{Poi}(\lambda)$
if
$$\mathbb{P}(X=k)=e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k=0,1,2,\ldots.$$

4. **Geometric random variable**
Let $p \in (0,1)$. Under the convention that X takes values in $\{1,2,\ldots\}$, we write $X \sim \mathrm{Geo}(p)$
if
$$\mathbb{P}(X=k)=p(1-p)^{k-1}, \qquad k=1,2,\ldots.$$

End of Lecture

Feedback and Corrections

These notes are prepared as lecture notes and may contain typographical errors, omissions, or mathematical mistakes. If you notice any such errors or have suggestions for improving the notes, please feel free to let me know.

Your feedback is greatly appreciated and will help improve future versions of these notes.

Report an error or suggest a correction

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